

The Heart's Story: What Influences Failure and Disease?

Tableau Public Link:

<https://public.tableau.com/app/profile/greg.charles/viz/ExploringFactorsInfluencingHeartFailureandHeartDisease/Story1>

Three Design Choices Based on Audience Persona:

My final data story in Tableau focuses on understanding the risk factors associated with heart disease and heart failure, built upon the synthetic dataset I created to ensure the inclusion of crucial variables. The design choices were heavily influenced by the needs and characteristics of my primary audience, **Akari Yamaguchi (General Resident)**. While the initial project proposal aimed to cater to three distinct personas, the complexity of visualizing the intricate relationships within the data led me to prioritize a design that would be most accessible and impactful for a general audience like Akari.

Here are three specific design choices I made with Akari Yamaguchi in mind:

- 1. Use of Simple and Intuitive Chart Types:** For Akari, who has limited medical knowledge and a busy schedule, clarity and ease of understanding are paramount. Therefore, I predominantly used straightforward chart types such as **bar charts** to compare prevalence across categories (e.g., heart failure rates among smokers vs. non-smokers), **scatter plots** to illustrate correlations between continuous variables (e.g., age and blood pressure, with heart failure as a color differentiator), and **highlight tables** to quickly identify high-risk groups based on combinations of factors. Complex chart types like box plots or advanced statistical visualizations were avoided to prevent overwhelming her. The language used in titles, labels, and annotations is also kept concise and free of jargon, aligning with her need for simple and direct information.
- 2. Emphasis on Color and Clear Visual Hierarchy:** To engage Akari and guide her through the information efficiently, I employed a deliberate color scheme and a clear visual hierarchy. **Consistent color coding** is used throughout the story to represent the presence or absence of heart disease and heart failure (e.g., a distinct, cautionary color for positive cases). **Larger fonts for key metrics and titles**, along with strategic use of white space, ensure that the most important information is immediately noticeable. The story unfolds in a logical flow, starting with the overall prevalence and then progressively drilling down into specific risk factors. This structured approach caters to Akari's need for information presented in an organized and digestible manner, allowing her to follow the narrative without feeling lost.
- 3. Interactive Personalized Risk Calculator:** A key design element tailored to Akari's goal of understanding her personal risk is the inclusion of an **interactive risk calculator**. This separate page allows users to input their own data for various risk factors (e.g., age, blood pressure, smoking status, BMI, presence of diabetes, etc.) and receive an immediate, estimated probability of developing heart failure, heart disease, and kidney disease. This feature directly addresses Akari's need for clear and relatable information about her potential health risks based on her individual circumstances, empowering her to have more informed conversations with her doctor and consider proactive lifestyle changes. The simple interface

with clear input fields and easily understandable probability outputs was designed to be accessible even for users with limited technical or medical knowledge.

Original Project Proposal and Reflection:

Project Proposal: Understanding and Visualizing Heart Disease Risk Factors: A Data-Driven Story for Everyone (Built with Tableau)

Executive Summary

Heart disease continues to be a significant public health concern worldwide. This project aims to utilize Tableau to analyze a comprehensive dataset focused on heart disease and related risk factors, including variables like **HighBP**, **HighChol**, **Smoker**, **Diabetes**, **PhysActivity**, **BMI**, **Age**, **Income**, and the direct outcome **HeartDiseaseorAttack**.

By creating an interactive and insightful Tableau Public workbook, the goal is to empower proactive heart health globally by providing tailored data-driven narratives for different audiences: general residents, healthcare professionals, and public health policymakers. The visualizations will highlight key risk factor prevalence, show the potential impact of modifiable behaviors and socioeconomic factors, and help inform targeted strategies for prevention and resource allocation worldwide.

Why This Matters: Improving Global Heart Health Through Data Insights

This project is driven by the need to address heart disease worldwide by turning data on risk factors into accessible and actionable knowledge for various stakeholders. The main objectives are to:

- **Help General Residents (Akari Yamaguchi):** Provide clear, relatable, and visually engaging information about their potential heart disease risks based on factors like blood pressure, cholesterol, smoking, activity levels, and BMI, encouraging them to take a proactive role in their health.
- **Support Healthcare Professionals (Dr. Emily Carter):** Offer visually rich and potentially interactive Tableau dashboards that enhance patient education, support early identification of at-risk individuals based on a broader range of indicators including healthcare access and socioeconomic factors, and facilitate informed clinical discussions in their practice.
- **Inform Public Health Policymakers (Yuta Sato):** Present data-backed insights into the prevalence and interplay of key risk factors, including socioeconomic determinants and access to care, within a representative dataset, enabling evidence-based decision-making for public health initiatives, resource allocation, and targeted interventions globally.

This project will use the power of Tableau to translate complex health data into meaningful stories that can contribute to a healthier world.

Specific Outcomes:

- **Interactive Risk Factor Prevalence Dashboard (Tableau):** Clearly visualize the proportion of individuals exhibiting various risk factors (e.g., **HighBP**, **HighChol**, **Smoker**, **Diabetes**, **LowPhysActivity**) within the dataset, allowing users to filter by age, sex, and income.
- **Comparative Health Outcome by Risk Factor (Tableau):** Illustrate how common **HeartDiseaseorAttack** is across different categories of key risk factors (e.g., smokers vs. non-smokers, those with high vs. normal blood pressure) using Tableau bar charts and highlighting significant differences.
- **Modifiable Risk Factor Impact Explorer (Tableau):** Specifically visualize the relationship between modifiable lifestyle factors (**Smoker**, **PhysActivity**, **Fruits**, **Veggies**, **HvyAlcoholConsump**, **BMI**) and the likelihood of **HeartDiseaseorAttack**, potentially showing the impact of positive changes.
- **Socioeconomic Disparities in Heart Disease (Tableau):** Analyze and present how often **HeartDiseaseorAttack** occurs in relation to **Education**, **Income**, **AnyHealthcare**, and **NoDocbcCost**, highlighting potential disparities within the population.
- **Age and Heart Disease Risk (Tableau):** Explore and visualize the correlation between **Age** and the prevalence of **HeartDiseaseorAttack**, potentially showing how risk increases with age and alongside other factors.
- **Policy-Relevant Insights Dashboard (Tableau):** Aggregate key findings into a dashboard specifically designed for policymakers, highlighting the most prevalent risk factors, socioeconomic disparities, and potential areas for public health interventions globally.

Who:

This data story, carefully put together in Tableau, is designed to resonate with three important groups of people worldwide:

- **Akari Yamaguchi (General Resident):** A health-conscious individual looking for clear, visual information about heart disease risks based on factors relevant to her daily life and socioeconomic context.
- **Dr. Emily Carter (Healthcare Professional):** A dedicated physician aiming to improve patient education and early risk identification by understanding how various risk factors interact, including socioeconomic influences and healthcare access.
- **Yuta Sato (Public Health Policymaker):** A representative of a governing body responsible for public health initiatives, seeking data-backed evidence on prevalent risk factors and socioeconomic disparities to inform targeted policy decisions and resource allocation for heart disease prevention.

Persona Document 1: Akari Yamaguchi (General Resident)

- **Role:** A busy working professional in her late 50s, concerned about maintaining good health as she gets older.
- **Age:** 58
- **Gender:** Female
- **Goals:**
 - Understand her potential risk factors for heart disease in a clear and simple way.
 - Learn about lifestyle changes she can make to improve her heart health, considering factors like access to healthy food options and opportunities for physical activity in her community.
 - Identify symptoms or indicators she should be aware of and discuss with her doctor.
 - Prefers information presented visually and without complex medical terminology.
- **Challenges and Needs:**
 - Limited medical knowledge.
 - Busy schedule makes it difficult to find time for extensive research.
 - Needs practical, actionable advice relevant to her daily life, considering factors like cost of healthy food and availability of safe spaces for exercise.
 - Seeks reassurance and motivation to prioritize her heart health.
- **Context:** Might come across the presentation online, at a local community health fair, or shared by a friend or family member.
- **(Optional) Other Relevant Notes:** Tries to stay active but finds it challenging due to work commitments. Concerned about the cost of healthcare and its impact on her ability to access preventive care.



Persona Document 2: Dr. Emily Carter (Healthcare Professional)

- **Role:** Family Physician with an interest in preventative medicine and patient education.
- **Age:** 45
- **Gender:** Female
- **Goals:**
 - Understand how data analysis can visually represent potential risk factors for heart disease to aid patient education in her diverse practice, considering factors like income, education, and access to healthcare.
 - Identify key indicators that patients with varying socioeconomic backgrounds should be aware of and discuss.
 - Evaluate the potential of data-driven insights to promote proactive health behaviors in her community, addressing barriers like lack of access to healthy food or safe exercise environments.
 - Wants accurate information presented clearly for translation into patient-friendly advice that acknowledges different life circumstances and cultural contexts.



- **Challenges and Needs:**
 - Limited time for deep data analysis.
 - Needs clear, concise, and patient-relevant visualizations that consider socioeconomic factors and healthcare access disparities.
 - Seeks practical takeaways for patient counseling that are sensitive to different levels of healthcare access and financial constraints.
- **Context:** Might view the presentation during a break at her clinic, shared by a colleague interested in public health, or accessed through a professional online platform.
- **(Optional) Other Relevant Notes:** Committed to addressing health inequities and empowering her patients from all backgrounds to manage their health effectively.

Persona Document 3: Yuta Sato (Public Health Policymaker)

- **Role:** Member of a national or regional Health and Welfare Committee, responsible for developing and implementing public health policies.
- **Age:** 48
- **Gender:** Male
- **Goals:**
 - Understand heart disease risk factor prevalence within a representative population to inform national or regional public health strategies.
 - Identify target demographics for interventions based on socioeconomic disparities and access to care, using data to justify resource allocation and program development.
 - Evaluate the potential impact of public health initiatives (e.g., smoking cessation programs, promotion of physical activity, access to affordable healthcare) on heart disease risk.
 - Needs clear, concise data-driven evidence presented in a way that supports policy recommendations and budget requests.
- **Challenges and Needs:**
 - Requires relevant data that highlights socioeconomic disparities and access to care, providing evidence for the need for targeted interventions.
 - Needs actionable insights that can be translated into concrete public health programs and policies at a larger scale.
 - Must balance public health needs with budget constraints and diverse community acceptance.
 - Appreciates clear visualizations and summaries that highlight key trends and inequalities in heart disease risk.
- **Context:** Might view the Tableau workbook as a briefing document for a committee meeting, during presentations on population health, or via an internal data portal.
- **(Optional) Other Relevant Notes:** Committed to improving population health and reducing health disparities through evidence-based policy and resource allocation.



What: A Comprehensive Dataset for Understanding Heart Disease Risk Worldwide (Analyzed in Tableau)

The dataset includes 253,681 records representing individual patients and has the following key columns: **HeartDiseaseorAttack** (binary outcome), **HighBP** (binary), **HighChol** (binary), **CholCheck** (binary), **BMI** (numerical), **Smoker** (binary), **Stroke** (binary), **Diabetes** (binary), **PhysActivity** (binary), **Fruits** (binary), **Veggies** (binary), **HvyAlcoholConsump** (binary), **AnyHealthcare** (binary), **NoDocbcCost** (binary), **GenHlth** (categorical), **MentHlth** (numerical), **PhysHlth** (numerical), **DiffWalk** (binary), **Sex** (binary), **Age** (numerical), **Education** (categorical), and **Income** (categorical). The source of this valuable dataset can be found on Kaggle:

<https://www.kaggle.com/datasets/kirollosashraf/heart-disease-csv>.

Why this data: A Useful Tool for Global Heart Disease Prevention

This dataset, analyzed using Tableau's strong capabilities, gives us a good way to understand the complex relationship between medical conditions, lifestyle choices, socioeconomic factors, and access to healthcare in relation to how common heart disease or heart attack is globally. Because it's so detailed, it allows for insights that are directly relevant to the health of people worldwide, the work of healthcare professionals, and the planning of public health policymakers trying to reduce the global impact of heart disease. Tableau's interactive features will let each stakeholder explore the data and find the patterns that are most important to them.

How: Delivering Actionable Insights Through Interactive Tableau Visualizations

This project will share its findings through a well-organized and interactive Tableau Public workbook, featuring:

- **Visually Compelling Dashboards (Tableau):** Each dashboard will be designed with a specific audience in mind, showing key insights about risk factor prevalence, outcomes, and disparities in a way that's easy to understand and engaging.
- **Interactive Filters and Parameters (Tableau):** Letting users explore the data based on things like age, sex, income, education, and individual risk factors, allowing for personalized risk assessment and focused analysis.
- **Clear and Concise Visualizations (Tableau):** Using good data visualization practices to make sure complex information is communicated effectively through bar charts, pie charts, scatter plots, and other appropriate chart types, with clear labels and notes.
- **Data-Driven Narrative (Tableau Story Points):** Guiding the audience through a logical flow of insights, starting with how common heart disease is overall, then looking at individual and combined risk factors, examining socioeconomic influences, and ending with actionable takeaways for individuals and public health efforts globally.
- **Calculated Fields (Tableau):** Using Tableau's analytical power to create new measures and group people in different ways, like creating combined risk scores or comparing heart disease rates across different income levels.
- **Actionable Recommendations (Integrated within Tableau):** Each part of the Tableau workbook will include clear and simple takeaways for the intended audience, suggesting

potential lifestyle changes for residents, things for doctors to discuss with patients about screening and advice, and areas for public health to focus on regarding access, education, and prevention programs worldwide.

Challenges:

- **Data Representativeness:** While aiming for global relevance, the specific group of people in the dataset might have different characteristics than the exact global population and health situations. This will be clearly mentioned in the Tableau workbook.
- **Self-Reported Data:** Keeping in mind that some of the data (like smoking history or how many fruits and veggies people eat) is likely self-reported and could have some inaccuracies. This will be noted in the limitations section.
- **Complexity of Global Health Determinants:** Recognizing that heart disease is influenced by many different factors that can vary a lot between regions and cultures. This dataset gives us a look at some important indicators.
- **Actionability of Insights Globally:** While providing data-driven observations, actually getting individuals and governments around the world to take action based on these insights will need more work and local efforts beyond this project.

Areas I Hope to Gain More Experience:

- Designing really interactive and easy-to-use Tableau dashboards for different non-technical audiences worldwide, especially in healthcare and public health, working with datasets that have lots of different types of information.
- Crafting compelling data stories within Tableau that really help people understand and motivate them to take action globally.
- Using Tableau's features to explore the connections between health outcomes and socioeconomic factors on a wider scale.
- Developing clear and practical recommendations within a Tableau project that are tailored to different user groups and situations worldwide.

Specific Questions for Peers:

- What Tableau techniques work best for showing the relationship between a yes/no outcome (**HeartDiseaseorAttack**) and a mix of yes/no, category, and number-based risk factors for a global audience?
- What's a good way to structure a Tableau workbook to really highlight socioeconomic differences in heart disease risk on a global level?
- Are there creative ways to visualize how multiple risk factors together increase the chances of heart disease using this kind of data?
- How can I make sure the recommendations I include in the Tableau project are actually useful and practical for the different residents, healthcare professionals, and policymakers around the world?

Reflection on Project Proposal

Looking back at my original project proposal, several aspects have evolved during the development process. Initially, I aimed to create a data story that catered to three distinct personas: a general resident (Akari Yamaguchi), a healthcare professional (Dr. Emily Carter), and a public health policymaker (Yuta Sato). However, as I delved deeper into visualizing the complex relationships within the heart disease and heart failure data (especially with the creation of a more comprehensive synthetic dataset), I realized that effectively addressing the nuanced needs of all three personas within a single, cohesive data story would be challenging without making it overly complicated or too broad for each individual.

The primary shift was the decision to **prioritize the general resident persona (Akari Yamaguchi)** in my final design. This focus allowed me to make deliberate choices regarding chart types, language, and interactivity that would maximize accessibility and understanding for someone without a medical background. The inclusion of the interactive risk calculator is a direct result of this prioritization, aiming to provide a personalized and actionable tool for Akari. While the broader insights presented can still be valuable for the other personas at a high level, the specific design elements are primarily tailored to Akari's needs for clear, visual, and personally relevant information.

I learned that while it's ambitious to cater to multiple distinct audiences simultaneously, sometimes a more focused approach yields a more impactful and effective communication for the primary intended user. Trying to accommodate everyone might dilute the message for each specific group. The process highlighted the importance of **iterative design and being flexible** when working with complex data. The initial plan was somewhat idealistic in its scope, and the practicalities of data visualization led to a refinement of the target audience to ensure a more successful final outcome. This experience has underscored the crucial role of the audience persona in driving specific design decisions and the need to adapt the project scope based on the data and the clarity of the intended message.

The Heart's Story: What Influences Failure and Disease?

Introduction

The Heart's Story: What Influences Failure and Disease?

Introduction	Patient Demographics	Cardiovascular Risk Prediction Dashboard	Age Distribution of Patients with Heart Disease and Heart	BMI and Cardiovascular Outcomes	Lifestyle factors	Smoking's Influence on Age-Related Heart Conditions	Impact of Physical Activity Levels on Cardiovascular Health	Cardiovascular Risk by Systolic and Diastolic Blood Pressure	Cardiovascular Health: Exploring Risk Factors and Medication	Lipid Triad and Cardiovascular Risk	Link Between Anemia, Kidney Disease, and Cardiovascular Health	M A C
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Introduction

Welcome to 'The Heart's Story,' an interactive data-driven exploration into the factors that may influence heart failure (HF) and heart disease (HD). For an optimum experience, I recommend viewing this analysis in full-screen mode.

In this analysis, we delve into a rich home-made synthetic dataset comprising 250,000 simulated patient records. Each record includes a variety of **demographic information** (like Age, Sex, Ethnicity), **lifestyle factors** (such as BMI, Smoking, Alcohol Consumption, Diet Quality, Physical Activity, Sleep Duration, Stress Level), **physiological measurements** (including Blood Pressure, Cholesterol levels, Triglycerides, Heart Rate, ECG Abnormality, HbA1c, Creatinine, ALT), **medical history** (Family History, Previous Heart Problems, Kidney Disease, Anemia, diagnoses of Hypertension, Diabetes, and High Cholesterol), and **medication usage**.

Our primary question is: What key factors and relationships within this simulated population appear to be associated with the prevalence of heart failure and heart disease?

Feel free to click on different categories and data points within the visualizations below to explore how the prevalence of heart conditions varies across different groups. See the differences firsthand and uncover potential trends and patterns that could shed light on the complex interplay of variables impacting cardiovascular health. Join us as we navigate this data landscape to understand the story our hearts might tell.

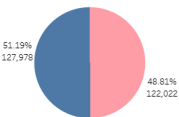
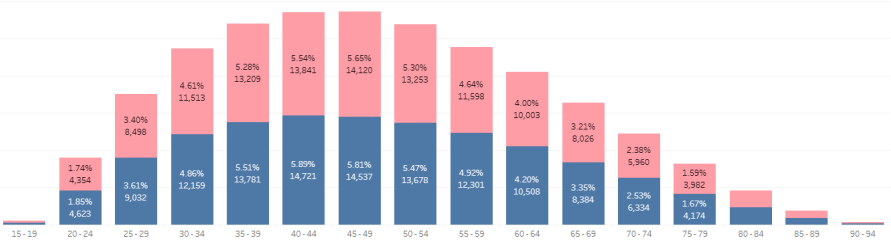
View on Tableau Public

Patient Demographics

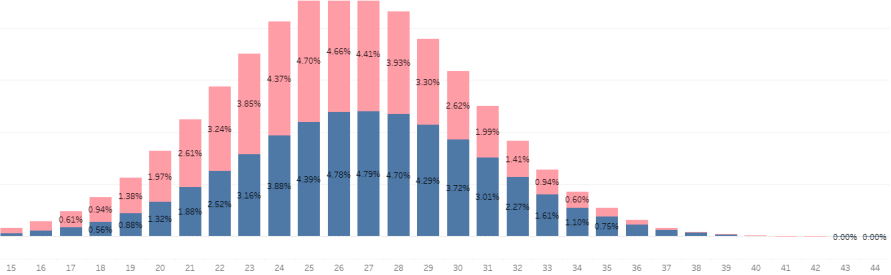
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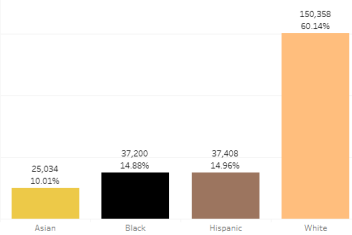
Patient Age Distribution



Patient BMI Distribution



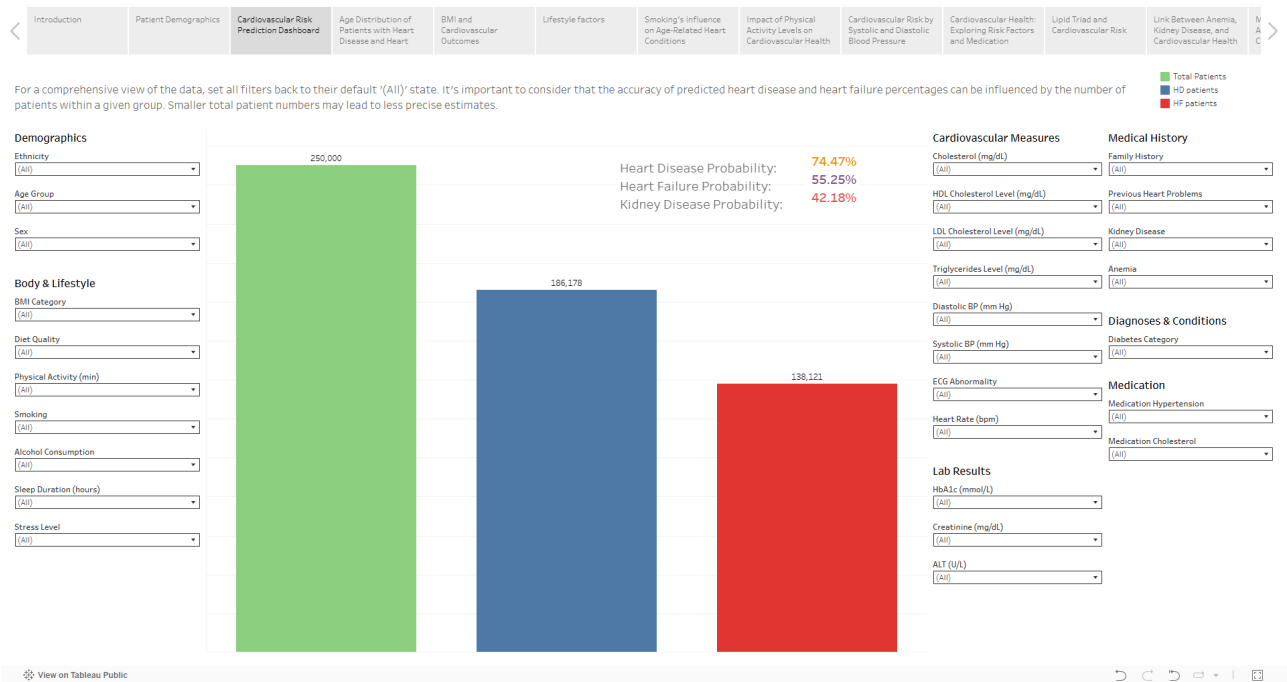
Ethnicity Distribution of Patients



View on Tableau Public

Cardiovascular Risk Prediction Dashboard

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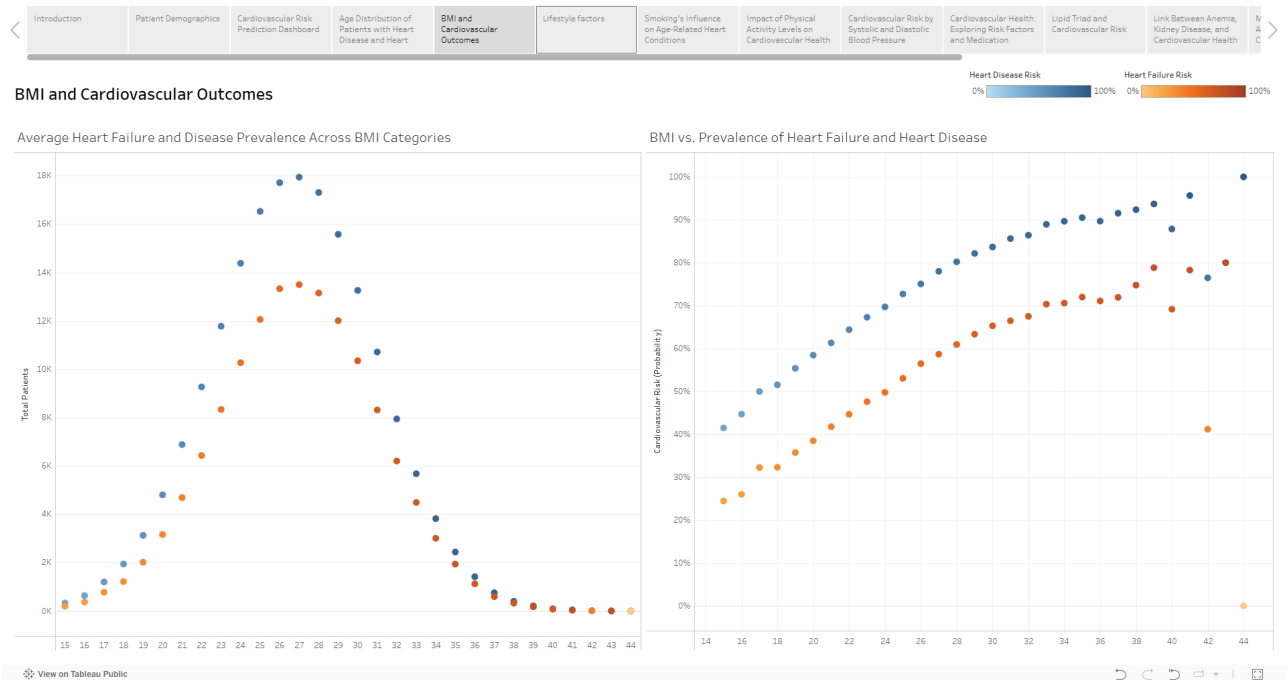
Age Distribution of Patients with Heart Disease and Heart Failure

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BMI and Cardiovascular Outcomes

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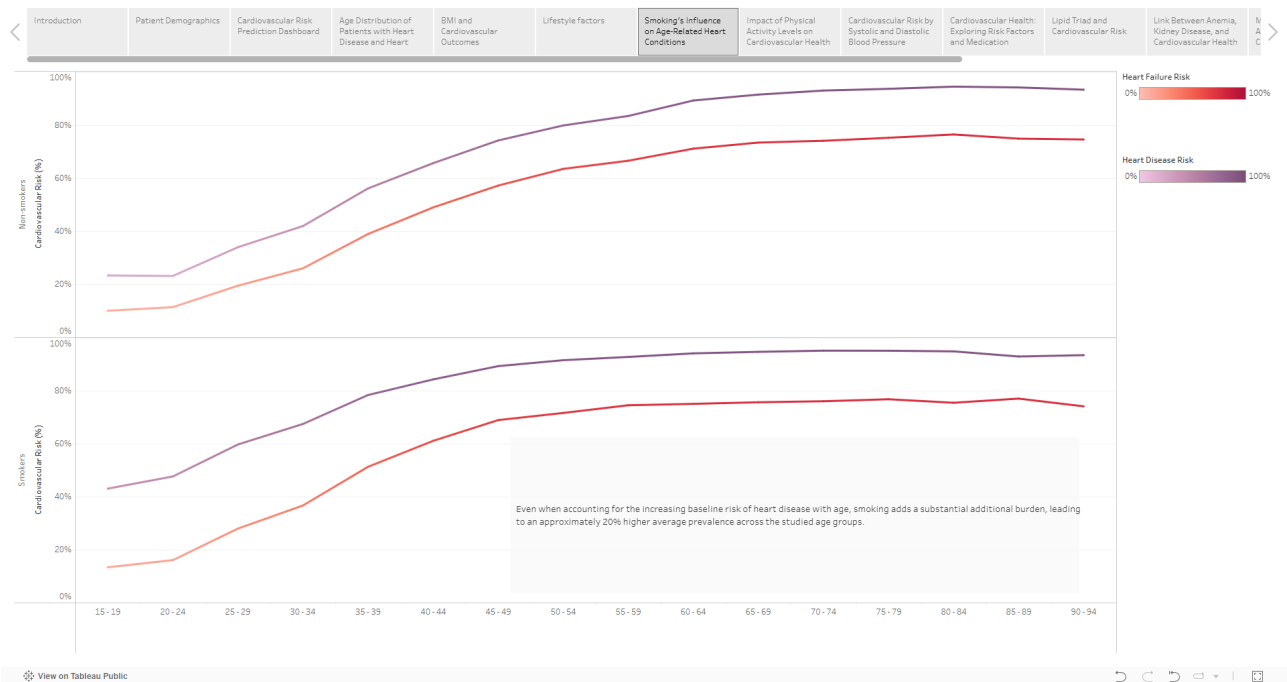
Lifestyle factors

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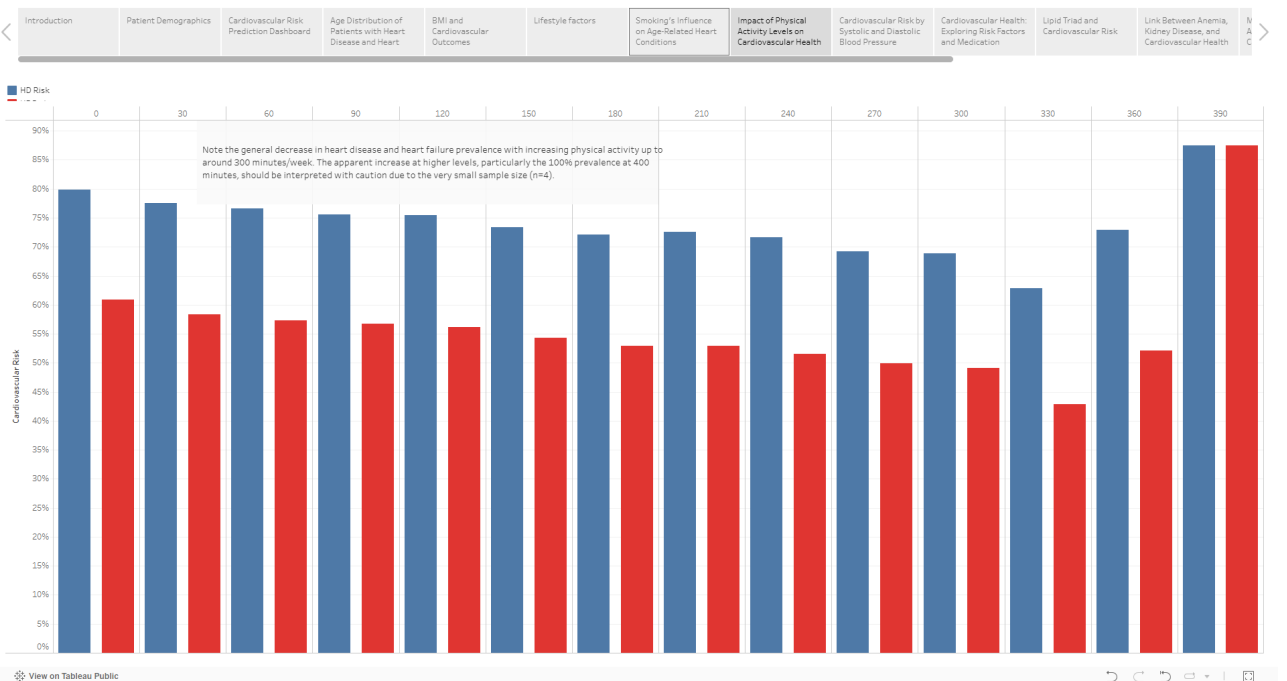
Smoking's Influence on Age-Related Heart Conditions

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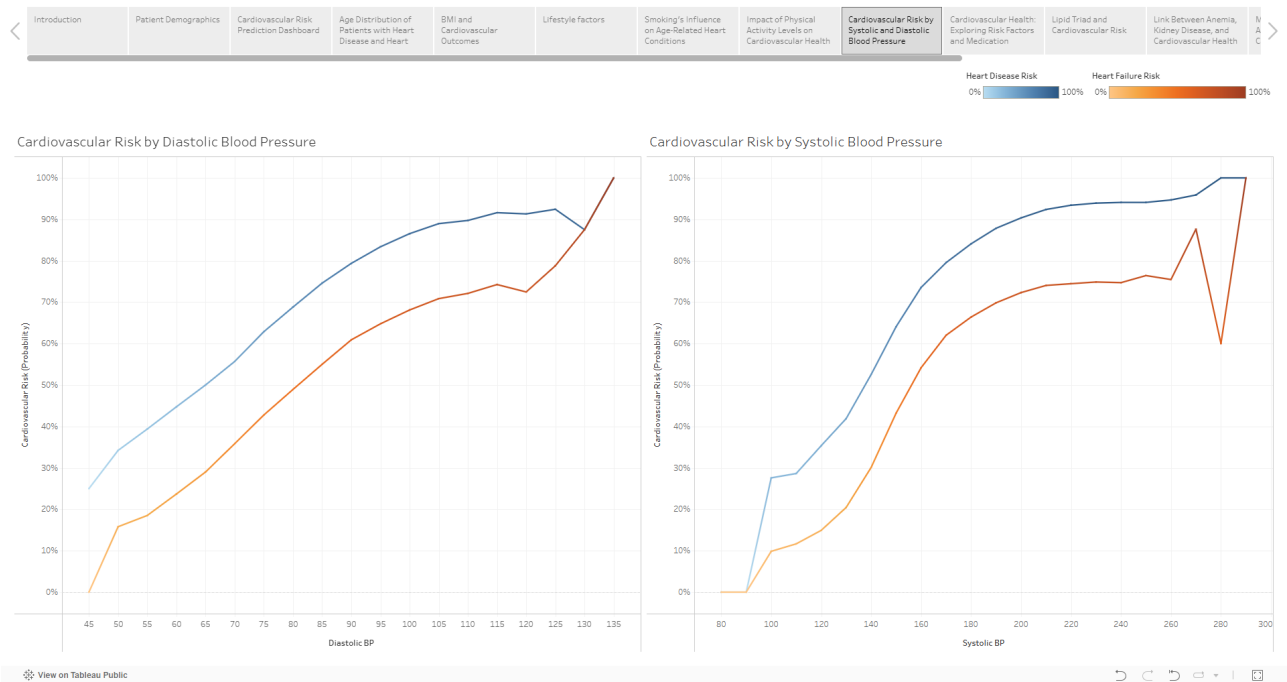
Impact of Physical Activity Levels on Cardiovascular Health

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Cardiovascular Risk by Systolic and Diastolic Blood Pressure

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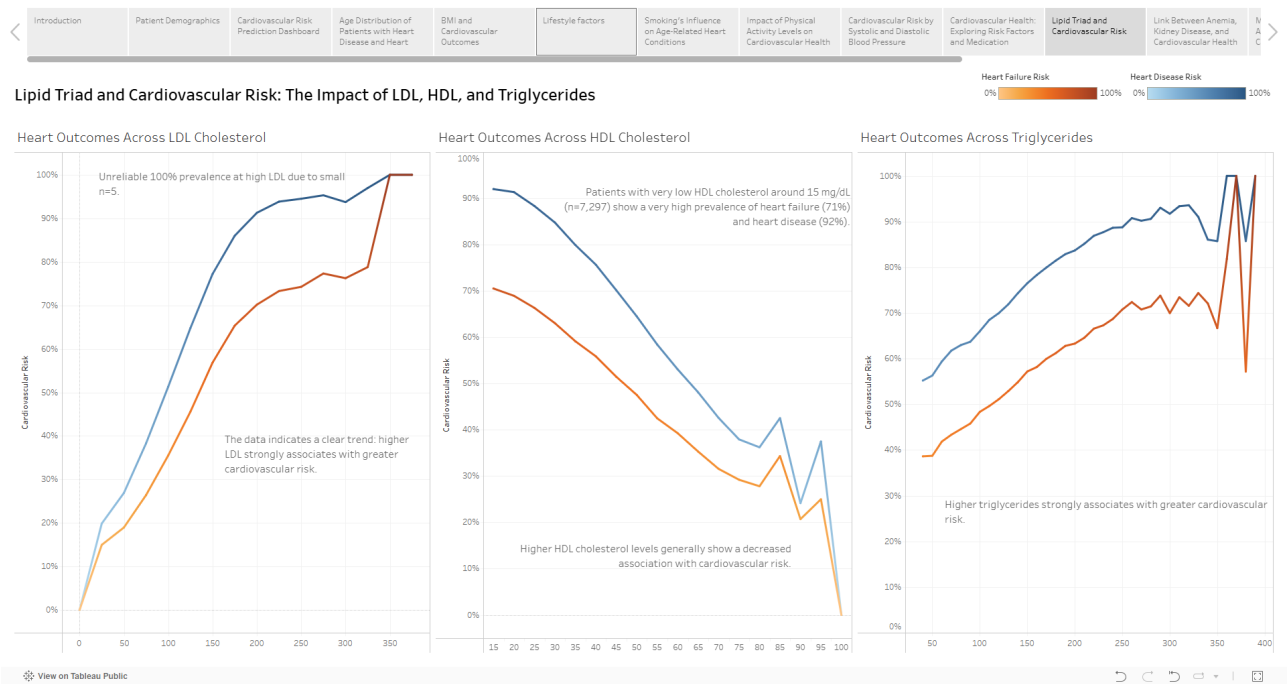
Cardiovascular Health: Exploring Risk Factors and Medication

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Lipid Triad and Cardiovascular Risk

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Link Between Anemia, Kidney Disease, and Cardiovascular Health

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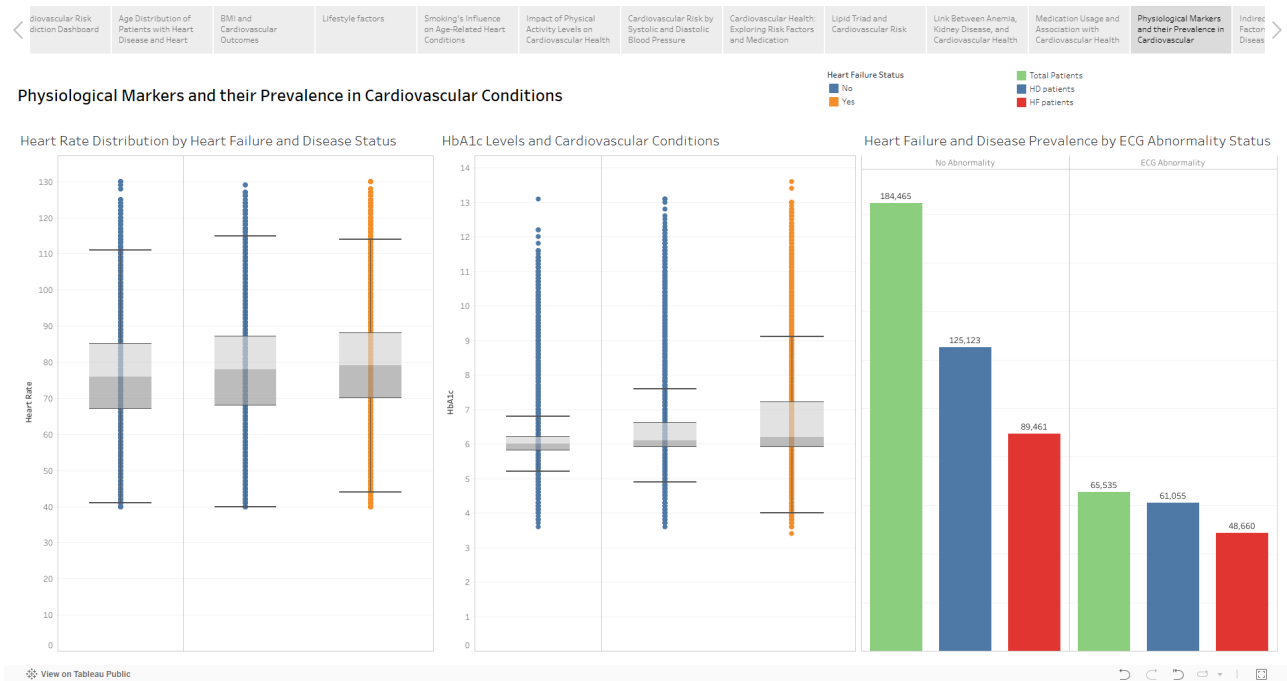
Medication Usage and Association with Cardiovascular Health

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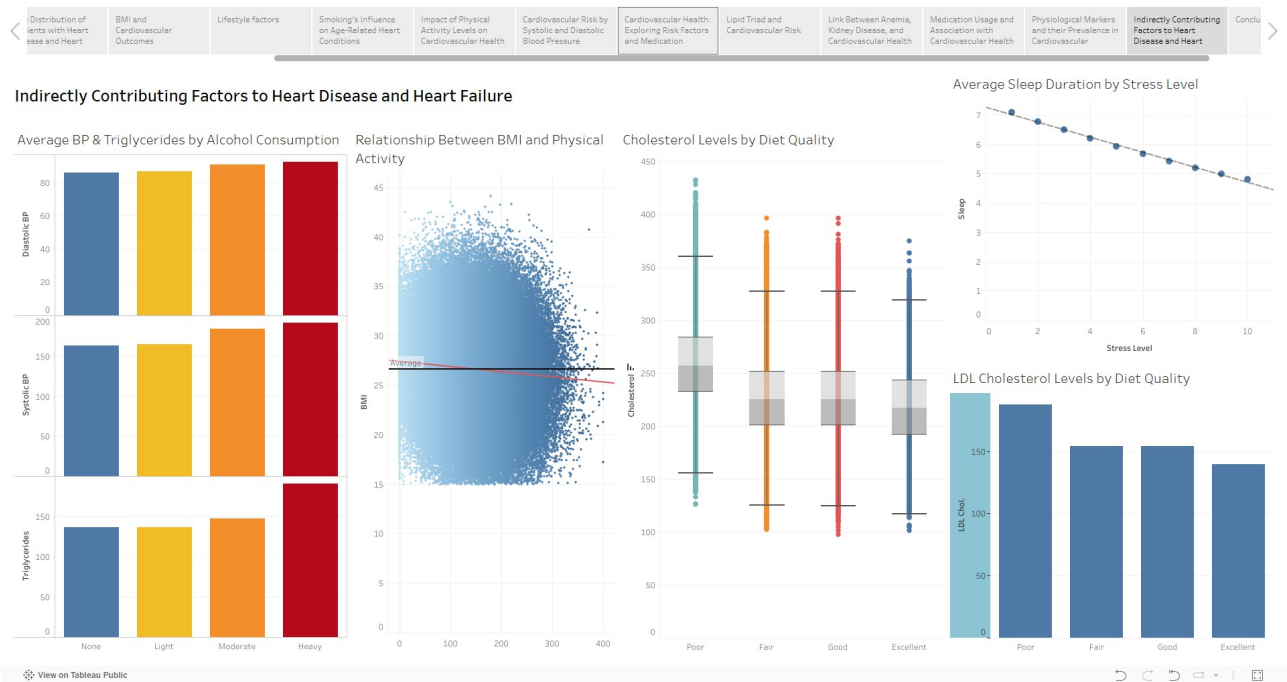
Physiological Markers and their Prevalence in Cardiovascular Conditions

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Indirectly Contributing Factors to Heart Disease and Heart Failure

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Conclusion

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